An Historical Income Panel for the Netherlands, 1850–1920

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Abstract: This paper presents the Historical Income Panel of the Netherlands (HIP-NL), a large-scale historical database that captures the incomes and key demographic characteristics of hundreds of thousands of households over the period 1850–1920. The data can be integrated with the Dutch microdata infrastructure, which is based on population registers and civil registers, allowing the welfare of household members to be studied from a multidimensional perspective while enriching these existing databases with key economic indicators. This paper describes the construction of the database, including a critical evaluation of the municipal tax registers on which it is based, the sample design, the data pipeline, the database's contents, and issues related to data harmonization and imputation. We also address the record linkage required to connect the tax data to external sources and introduce case studies on income inequality and migration to demonstrate the database's potential. Future avenues for research are discussed in the last section.

Keywords: Historical Income; Household Taxation; Historical Database; Income Inequality; The Netherlands

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Introduction

There has been growing interest among economic and social historians in questions of welfare, economic inequality, and social mobility. The work of scholars such as Piketty (2014), Alfani (2021), and Scheidel (2017) has placed the long-term development of economic inequality firmly on the research agenda. Research on social mobility has similarly received a major impetus from the work of Clark and Cummins (2014) and Chetty et al. (2014), among others. These questions have become all the more relevant in light of current societal concerns about inequality and social mobility, as well as growing recognition of the multidimensional nature of welfare and inequality, encompassing differences in education, health, longevity, and life opportunities in the broadest sense (see Deaton, 2024, and the articles collected in that special issue).

Addressing these questions over the long run requires high-quality, representative data. Research on societies prior to WWI relies to a considerable extent on observations from a limited number of places and periods (Alfani, 2021; Atkinson et al. 2011). There is a need for data that allow economic resources such as income and wealth to be measured for individuals and households, repeatedly over time and across generations. Such longitudinal and intergenerational data are essential for studying not only the distribution of economic resources, but also their persistence and transmission over the life course and between generations. Ideally, these data should also be linkable to other dimensions of welfare, including life expectancy, health and morbidity, and gender inequality, as well as to broader contextual indicators such as GDP per capita (Van Zanden et al., 2014; Rijpma et al., 2024).

Collecting and linking such data is both time- and resource-intensive. Consequently, much of the literature has relied on imperfect measures of economic welfare and inequality, such as occupation in studies of social mobility and social tables in inequality research. Such measures can conceal substantial variation in economic resources within occupational groups or households. A farmer, for example, could be a poor subsistence farmer or one of the wealthiest members of a community; the same applies to merchants, traders, and entrepreneurs. Reliance on such proxies can therefore introduce substantial measurement error and obscure important degrees of inequality (Modalsli 2015; Von Fintel et al. 2023; Ward 2023). When possible, it is preferable to measure income directly at the individual or household level.

The Netherlands illustrates the potential and the limitations of existing historical data. Historical demographers have used large-scale databases such as the Historical Sample of the Netherlands (HSN) and the digitised civil registries (LINKS) to reconstruct demographic behaviour for the late nineteenth and early twentieth centuries (Mandemakers, 2000; Mandemakers, et al. 2023; Mandemakers & Kok 2023), a crucial period encompassing the demographic transition, industrialization, the emergence of the welfare state, and major political transformations. The economic dimension, however, has not yet benefited from similarly high-resolution data. The Netherlands has world-class aggregate statistics, including the nineteenth-century time series underlying the Dutch

historical GDP estimates (Smits et al., 2014; Van Zanden and Van Riel, 2004; Van Riel, 2021). From 1915 onwards, inequality has also been reconstructed using national income tax tabulations, most notably by Salverda and Atkinson (2007). For the period before 1917, tentative estimates have been produced by Soltow and Van Zanden (1998), drawing on a wide range of proxy measures. What remains lacking, however, are systematically collected microdata that allow the economic dimension of welfare to be studied at the individual or household level for this crucial period.

This gap is not unique to the Netherlands. Internationally, a similar picture emerges, as large-scale demographic databases are now the state of the art in terms of coverage, richness, and their ability to study long-term change at the individual and household level (Edvinsson et al., 2023; Mandemakers et al. 2023; Ruggles, 2012). The economic dimension, however, is generally much less well represented, with occupation often serving as the principal indicator of socioeconomic position. Historical census microdata in particular have become an important resource for historians and social scientists, but they generally contain little information on economic welfare beyond occupations. The United States after 1940 is a notable exception, when census data include information on individual and household income. One important European exception is Sweden, where Bengtsson and Molinder (2024; 2025) analyse individual-level income data for the 1870–1950 period. Similar efforts are also being undertaken for Italy (see the other contributions to this special issue).

Against this background, this paper introduces the Historical Income Panel of the Netherlands (HIP-NL), a large-scale historical database designed to address these gaps. HIP-NL links historical municipal tax records to population registers and vital registration records to reconstruct household income and tax-related variables and provides key demographic information for hundreds of thousands of households between 1850 and 1920. The database can be integrated with the broader Dutch microdata infrastructure based on civil registration and population registers (Raad et al., 2020; Mandemakers, 2000; Van den Berg et al., 2020). This makes it possible to study the economic dimension of welfare alongside demographic and other life-course characteristics, while at the same time enriching existing demographic databases with detailed information on income, tax and other economic characteristics.

This paper introduces the HIP-NL database and its potential for research on historical income and inequality. First, we discuss the municipal income tax registers (*hoofdelijke omslag*) on which the database is based, explaining how the source material was sampled and collected using a combination of manual and automated processes. We then describe the data pipeline and the record-linkage procedures used to connect the tax records to external sources, including population registers and vital registration records. Given the sparsity and heterogeneity of the source material, record linkage presents a particular challenge. After describing the resulting dataset, we demonstrate its potential through two empirical case studies. The first examines the development of income inequality in the Netherlands between 1860 and 1920, placing our estimates in the context of existing estimates for later periods. The second investigates income differences between

migrants and natives in the city of Utrecht. The findings illustrate the analytical potential of HIP-NL. In particular, the inequality case study shows that income inequality in the Netherlands was high and rising in the three decades before WWI. The migration case study reveals an income premium for migrants, particularly those who originated from an urban environment and who moved over longer distances, in line with work emphasizing the positive selection of long-distance migrants. These case studies illustrate some of the potentials of the data. Future research avenues are outlined in the last section of this paper.

The Municipal Income Tax Registers

The core sources for the Historical Income Panel for the Netherlands are municipal income tax registers. In the first decades of the nineteenth century, municipal taxation was divided into local excise duties and taxes on assets – such as housing, furniture, and personal property – and land. The Municipalities Act (*Gemeentewe*t) of 1851 provided the legal framework that allowed municipalities to levy local income taxes. Following the abolition of local excise duties in 1865 (Staatsblad no. 79), local income taxation gained further importance (Smit 2002: 180–81; cf. Aidt and Jensen 2009: 162, 173, who argue that only local poll taxes or wealth taxes existed), becoming the dominant source of municipal revenue by the beginning of the twentieth century (Klep et al. 1982: 7).

The registers documenting municipal income taxation are commonly referred to as *Hoofdelijke Omslagen* (HO). Although HO taxes existed from the early nineteenth century, the number of municipal tax registers increased substantially following the legislative changes of 1851 and 1865 (see Figure 2). From 1920 onward, municipal income taxation became the responsibility of the national government, explaining the sharp decline in the number of HO tax registers from 1921 (Klep et al. 1982: 8).

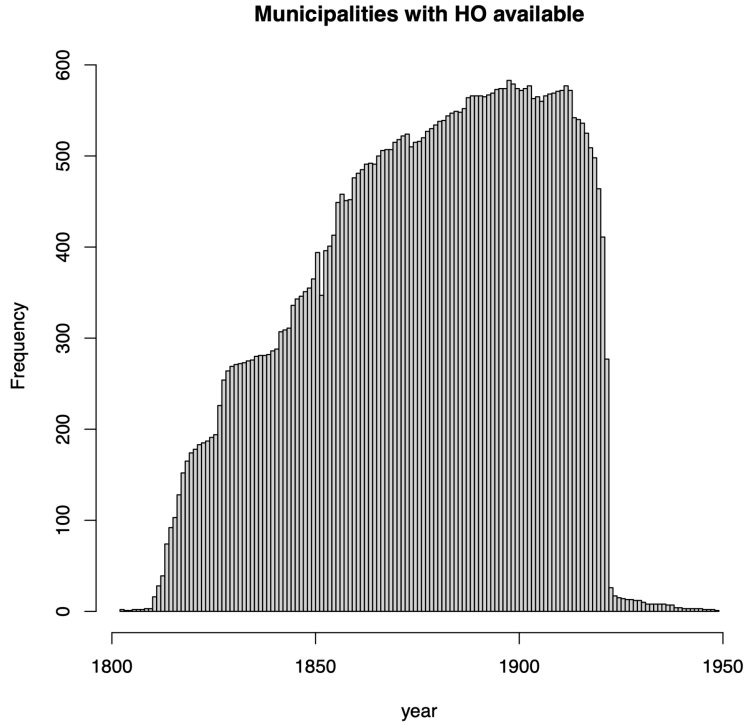


Figure 2: Number of municipal archives holding HO tax registers, 1800–1950.

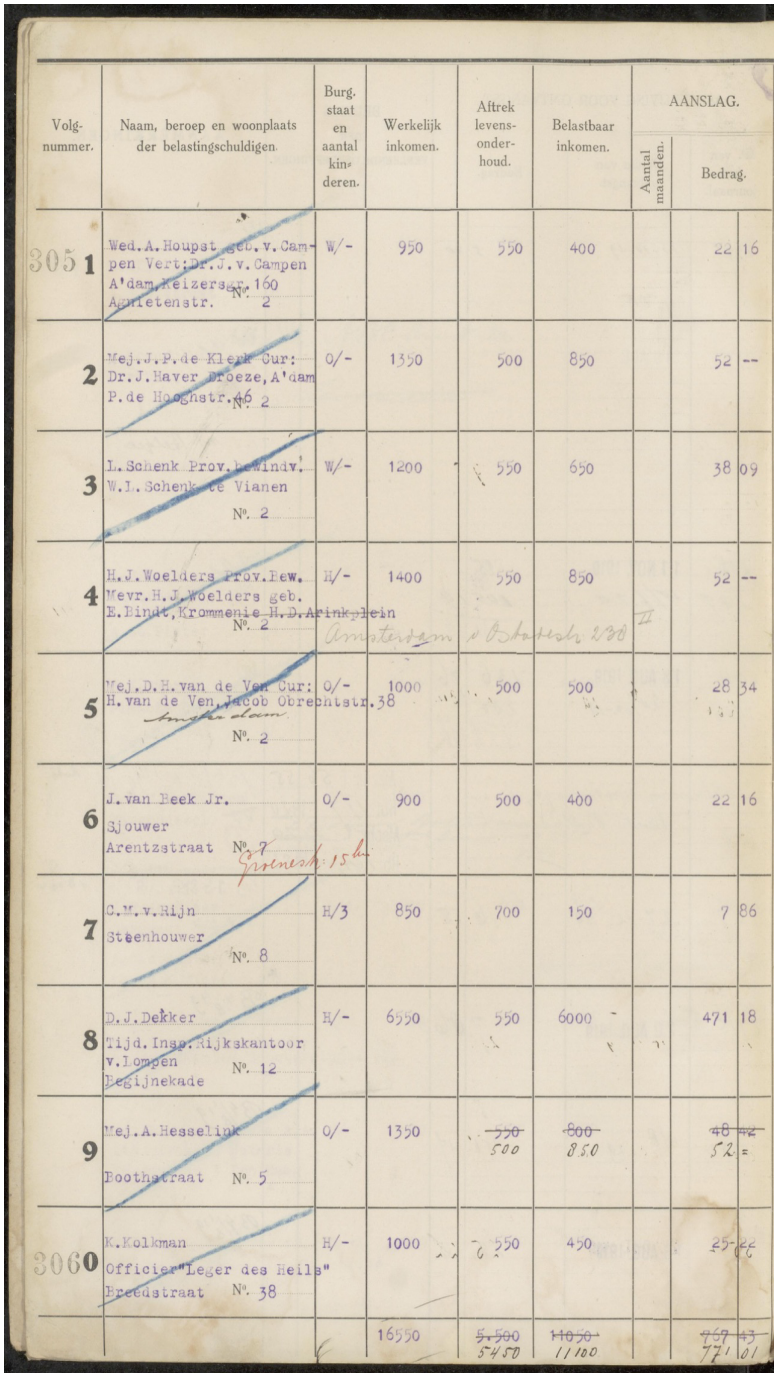
The municipal HO tax registers provide a unique opportunity to study historical income at the micro level. Other national taxes in the Netherlands either did not target income (the *personele belasting* primarily taxed residential rental values) or their records have been lost (the 1893 business and professional tax and the 1914/15 personal income tax; see Fritschy 1997; Smit 2002). The HO tax, by contrast, was explicitly intended to tax income, even if the concept of income was underdeveloped in the early decades of the tax. Initially, municipalities relied on civil servants or committees of local notables to assess incomes; by the twentieth century, filing tax returns had become common. The registers were made public to ensure the accuracy of the returns (Vording 2023). This practice was probably also one of the reasons why substantial numbers of these registers have survived.

The exact implementation of the HO tax was left to the municipalities, including the decision whether to levy it at all. Moreover, there was considerable variation in tax procedures and reporting practices, making the processing and harmonisation of HO tax records challenging. Nevertheless, some common characteristics can be identified.

The registers recorded tax assessments based on personal or household annual income, which was often estimated by municipal officials or committees of local notables (Vording 2023: 14–16). Later in the period, incomes were often self-reported. Entries in the *hoofdelijke omslagen* were recorded under individuals' names, and it was usually the head of the household who was listed. This often means that the HO tax can be treated as

a household tax, although in some municipalities and periods, individuals earning an independent income within the household were listed separately.

The income information recorded in the registers varied between municipalities and years. At a minimum, the registers reported the taxes levied, the name of the head of the household, and basic location information, often including the street and house number or the neighbourhood in which the household resided. In other municipalities, particularly in later years, the *hoofdelijke omslagen* included income measures, demographic information relevant to tax deductions – such as the number of children or marital status – and occupations. The income measures recorded varied and could include gross income, taxable income (gross income minus deductions), or income figures that were not further specified. Income classes were also frequently used. The example of Utrecht in 1919 (Figure 3 below) illustrates a register containing detailed information on location, demographic characteristics, occupation and, crucially, income. Most tax registers reported at least names, taxes levied and income information.



Volg-nummer.	Naam, beroep en woonplaats der belastingschuldigen.	Burg. staat en aantal kinderen.	Werkelijk inkomen.	Aftrek levens-onderhoud.	Belastbaar inkomen.	AANSLAG. Aantal maanden.	AANSLAG. Bedrag.	
3051	Wed. A. Houpst geb. v. Campen Vert; Dr. J. v. Campen A'dam, Keizersgr. 160 Adrietenstr. N°. 2	W/-	950	550	400		22	16
2	Mej. J. P. de Klerk Cur: Dr. J. Haver Utrecht, A'dam P. de Hooghstr. 46 N°. 2	O/-	1350	500	850		52	--
3	L. Schenk Prov. Ie Windv. W. L. Schenk te Vianen N°. 2	W/-	1200	550	650		38	09
4	H. J. Woelders Prov. Rew. Mevr. H. J. Woelders geb. E. Bindt, Krommenie H. D. Arinkplein N°. 2	H/-	1400	550	850	II	52	--
5	Mej. D. H. van de Ven Cur: H. van de Ven, Jacob Obrechtstr. 38 N°. 2	O/-	1000	500	500		28	34
6	J. van Beek Jr. Sjouwer Arentzstraat N°. 7	O/-	900	500	400		22	16
7	C. M. v. Rijn Steenhouwer N°. 8	H/3	850	700	150		7	86
8	D. J. Dekker Tijd. Insp. Rijkskantoor v. Lompen Begijnekade N°. 12	H/-	6550	550	6000		471	18
9	Mej. A. Hesselink Boothstraat N°. 5	O/-	1350	~~550~~ 500	~~800~~ 850		~~48~~ 52	~~42~~ =
3060	K. Kolkman Officier "Leger des Heils" Breedstraat N°. 38	H/-	1000	550	450		~~25~~	~~66~~
			16550	~~5.500~~ 5450	~~11050~~ 11100		~~769~~ 771	~~43~~ 01

Figure 3: HO Tax register for Utrecht in 1919.

Incomes were commonly reported by taxpayers themselves, especially in later years. In earlier years, municipal officials also estimated incomes, which were recorded as 'suspected income' ("*vermoed inkomen*"). Tax deductions were often applied, allowing a basic subsistence amount to be subtracted from gross income. In addition, a deduction could be granted for each child below a specified age. The most straightforward implementation of this system involved reporting gross income, the deductions in guilders

(fl.), and taxable income. In some cases, the registers do not make clear whether deductions were applied.

The deduction system itself meant that many HO taxes were progressive, in the sense that low-income households paid a smaller proportion of their gross income in tax than high-income households. In addition, progressive tax rates were applied on top of the deduction system in approximately one in ten municipalities for which we have information on taxable income. This practice became more common after 1900, as greater degrees of progressivity were permitted (Smit 2002: 180, 268). For 80 percent of the households for which taxable income can be observed, tax rates on taxable income ranged from 1.0 to 6.2 percent, while tax rates on gross income ranged from 0.4 to 3.5 percent.

In earlier tax years, particularly in the 1850s and 1860s, the registers often report only the taxes due, without providing the underlying income figures or deductions. Determining whether these early taxes were also progressive is one of the key challenges in using the HO tax registers to study the entire period.

The individuals recorded in the registers do not represent the entire population of the respective municipalities. First, the male head of household was often registered as the taxpayer, making this the most common form of registration. Married women earning an income were rarely registered in the HO tax registers; most women appearing in the tax records were household heads because they were widowed or unmarried. The tax unit in the HO was often the household, and married women were only very rarely registered separately. One deviation from this pattern is that older, often unmarried, children could be registered as independent taxpayers, separately from their parents. This occurred particularly in cases where unmarried sons lived at the same address as their widowed mothers. In rare cases, married women were also registered separately. Another exception to the household-based approach concerns siblings who earned separate incomes and therefore paid taxes individually despite living at the same address.

Institutional households are also sometimes found in the HO tax registers. These could take the form of a larger household in which several people earned an income but paid income tax jointly, as in an abbey or convent where the head of the household was taxed. Alternatively, an institutional household could consist of a larger group of individually taxed people registered at the same address. An example is the group of women registered as individual taxpayers at a psychiatric home and hospital for women in Utrecht, an institution that catered particularly to women from wealthy families.

Second, an income threshold applied, excluding households whose incomes were at or below subsistence level from taxation. The threshold varied by municipality, ranging from fl. 200 to fl. 500. It was generally lower in earlier periods and in rural areas, and higher in cities. Consequently, a sizable share of the population with incomes below this subsistence level was excluded from the tax registers. To correct for this, we establish a reference population using household counts from Dutch censuses and municipal reports, estimating these from total population figures where necessary (Boonstra et al. 2003).

These reference populations can be used in bounding exercises to assess the robustness of microdata estimates. Where high-quality population registers are available, non-taxed individuals can also be identified, allowing researchers to estimate incomes for all tax units in a municipality. The linking process and the requirements for population registers are discussed below.

Sample Design

Our sample of *Hoofdelijke Omslag* tax records spans more than seventy years (1849–1921). Sampling began in 1851 and continued at ten-year intervals from 1859 onwards to coincide with the Dutch censuses. We aimed for a sample of municipalities that would cover 10% of the Dutch population. We sampled municipalities for three reasons. First, the HO tax registers are held in municipal archives, and sampling individuals would have required us to visit every archive. Second, sampling municipalities allows us to establish a reference population. Third, sampling within municipalities allows us to follow individuals who did not move over time, turning the data into a panel.

The original sample was stratified by province and population size (see Figure 4). Municipalities with substantial populations were selected manually to ensure the availability of HO tax registers. The cities of Utrecht, Leiden, Nijmegen, Enschede, and Eindhoven were included for this reason. Amsterdam was excluded because, when the sample was designed, its population registers had not yet been digitised, limiting the extent to which these data could be integrated into the wider Dutch microdata infrastructure.

Beyond these larger, manually selected cities, municipalities were selected at random within provinces and municipality-size strata (fewer than 5,000; 5,000–10,000; and 10,000–50,000 inhabitants), following a preliminary check of the digital availability of population registers for the respective municipalities. This check meant that the province of Groningen and most of Overijssel had to be excluded from the sample because no digitised population registers were available. Access to population registers is important for the further use of the income panel. The tax registers provide names and addresses or other location information, whereas the population registers contain names, addresses or locations, and years of birth. The latter information is necessary to identify individuals uniquely and to enable further record linkage.

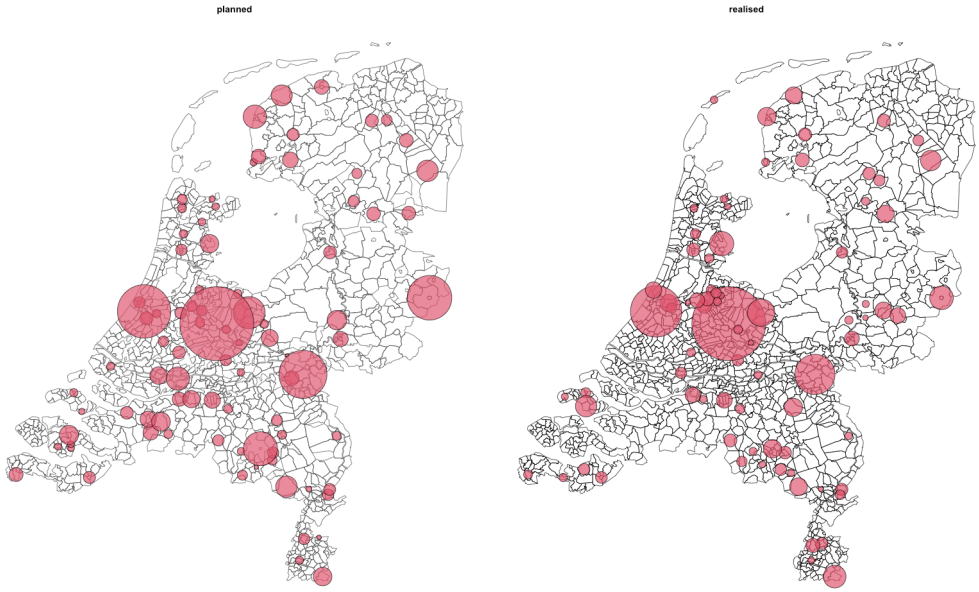


Figure 4: Municipalities in the planned (left) and realised (right) HIP-NL sample. Circles are proportional to the municipality's population size in 1899.

Data Pipeline

Data Extraction

In this section, we discuss the data pipeline used to create a coherent database from the separate tax registers. For full details, we refer to the working paper version of this paper, which provides a comprehensive account of the record linkage procedure (XXXX et al. 2026), as well as to a separate paper describing our data extraction workflow based on Vision Language Models (XXXX and XXXX 2026).

Data collection involved a combination of manual transcription by research assistants (RAs) and automated extraction using OCR and VLM models. Most of the data, particularly from small villages, were collected manually by RAs. They were instructed to scan the registers for a sampled municipality for every tenth year: 1859, 1869, 1879, 1889, 1899, 1909, and 1919. If a register was unavailable for a particular year, the nearest available year was selected instead. If the HO registers were unavailable in a municipal archive, an alternative municipality of a similar population size in the same province was selected.

We developed an entry form that followed the structure of a typical HO tax register as closely as possible while also facilitating rapid data entry. The information transcribed from the registers included, of course, income (gross, taxable, or unspecified), deductions, and taxes due. In addition, we recorded surnames, initials or first names, addresses, neighbourhoods, and titles of the head of the household or the individual liable for the tax. Information on children and marital status was also recorded. In some

municipalities, occupations were reported as well. We can often infer sex from names, titles such as widow, Ms, or Mrs, or maiden names. In some municipalities, the number of months for which the tax was due was also reported for residents who had not lived in the municipality for the entire year. Changes of address were also recorded. Finally, although information on the actual payment of the tax was usually included in the registers, we did not include it in our transcription. A full list of the variables recorded from the HO tax registers is provided in Appendix A.

In larger municipalities, the HO tax registers were printed and, towards the end of the period, also typewritten. Initially, we processed these registers using fine-tuned versions of EasyOCR. Although these produced sufficiently accurate results, considerable work remained to reconstruct the original tables from the OCR output, which consisted of text and bounding boxes. In addition, each register required a new fine-tuning step to achieve sufficiently accurate results.

For these reasons, we decided to experiment with more efficient methods of extracting data from the scans. We first experimented with an "OCR-free" approach called DONUT (Kim et al. 2022). Unlike traditional OCR models, which recognise individual characters or words in separate text regions, this model processes the scan as a whole. It is an example of a Vision Language Model (VLM), which combines image understanding with language modelling. Such models can be trained to answer questions about an image, transcribe text directly from a page, or even extract a table from an image and return it in tabular form. We trained DONUT to extract the required tables directly from the scans. This approach was applied to the Leiden and Nijmegen records.

Although the DONUT approach was an improvement over traditional OCR methods, fine-tuning a VLM remained time-consuming, and separate fine-tuning appeared to be necessary for each municipality. We suspected that, by late 2024, general-purpose models might already have surpassed our case-specific approach based on a model developed in 2022. An exploration of several models showed that the Gemini models (Gemini Team 2023) performed very well on our tasks. They required only a well-designed prompt describing the information to be extracted from the scans and, in some cases, a few examples of the desired output (few-shot prompting).

Using Google's generative AI API, we employed these models to extract data from the printed records for Enschede, the typewritten records for Utrecht, and—most notably—the handwritten records for Utrecht in 1899. Given the high quality of the output, we also decided to reprocess the printed records for Leiden and Nijmegen using this approach. The model output was always checked by the RAs, but this process was considerably less time-consuming than full manual transcription. We believe this approach holds great promise for future digitisation projects involving structured historical sources, except for the most challenging handwritten materials. We describe this approach in greater detail in a separate paper (XXXX and XXXX 2026).

Record Linkage

Linkage of households to the population registers is a crucial step in the construction of the HIP-NL dataset. It enriches the sparse tax data with demographic information, such as dates and places of birth and, in some cases, marital status, household composition, and address. By linking households to a common observation in the population registers, we can also connect household observations in the tax registers over time, effectively turning our tax data into a panel. Finally, this linkage allows our data to be integrated into the Dutch historical microdata infrastructure, particularly LINKS (Raad et al. 2020) and the HSN (Mandemakers 2000).

Our primary aim is to link tax observations to the population registers and, in some cases, to their successors, the family cards. The population registers record all residents of a municipality over a given period, reducing the search space for individuals compared with the entire Dutch population and making it possible to identify individuals recorded in the HO tax registers. In our experience, households recorded in the HO tax registers can almost always be found in the full population registers, as the combination of names and addresses in the HO tax registers generally allows them to be identified uniquely. The dates and places of birth obtained from the population registers then enable us to identify taxed individuals uniquely in other databases and sources.

However, automatically linking these sources is challenging for two main reasons. First, we need to allow for variation and minor errors in the records. This is a familiar record linkage problem (Christen 2012) that has been addressed successfully in social, demographic, and economic history using a variety of approaches (Rijpma et al. 2020; Abramitzky 2021; Raad et al. 2020). The second, and more substantial, challenge is the extreme heterogeneity and sparsity of the data. The information contained in the tax registers varies from one municipality to another: full names, occupations, street names, and house numbers are not consistently reported. Furthermore, we rely on volunteer transcriptions of the population registers, and the information transcribed also varies across municipalities. Some transcriptions contain only names and dates and places of birth, meaning that matching can rely only on names and age at the time of the HO tax observation. In other cases, address or occupation information is also available.

We retrieved the necessary population registers, stored as XML data, from archival repositories and cross-referenced these data against listings scraped from archival websites. We then retrieved any missing registers, which in some cases required us to scrape the archival websites directly. Once we had assembled a complete dataset, we standardised column names and register data. We also created unique identifiers for pages to group individuals appearing at the same address. In addition, we enriched the data for our linkage pipeline by training a model to classify whether the order of individuals in our transcriptions was likely to correspond to their order in the original population registers. We also trained a model to predict the relationships between individuals and the household

head within population register pages, using Conditional Random Field (CRF) models (Lafferty, McCallum, and Pereira 2001).[1]

With both the HO tax data and the population registers in place, the first step in the linkage procedure is blocking. For each person in the HO tax data (the 'query'), we identify a subset of 15 plausible candidates from the population registers, forming a candidate block. Blocking is based on a combined string containing the name and, where available in both the HO tax data and the population registers, the occupation and/or address (e.g. 'J. de Vries | Bakker | Oudegracht 123'). We then use a sentence encoder to encode each string into a 768-dimensional vector space (the pre-trained multilingual E5 model; Wang et al. 2024) and use FAISS (Facebook AI Similarity Search; Douze et al. 2024) to store the encoded candidates and retrieve the closest matches.

This semantic blocking approach offers significant advantages over conventional string-distance methods. Blocking based on embeddings allows us to capture semantic similarity between records, even when there are variations in spelling (Maria and Marie), abbreviations (straat and str.), or the use of synonyms or diminutives (Willem and Wim). Semantic similarity proved particularly useful for occupations: for example, 'jurist' and 'advocaat' have very similar embeddings despite having a large string distance. This approach leads to more effective and robust candidate-link generation, particularly for noisy historical data. Furthermore, it is computationally efficient because each string needs to be encoded only once, after which the resulting numerical representations can be searched efficiently using FAISS.

The blocking procedure consisted of two rounds. Primary blocking was based on names only. Secondary blocking added occupation and/or address information to the name strings of the candidates retained after primary blocking, where these data were available in both the HO tax data and the population registers. When constructing the name strings, we prioritised the following options, in order of preference: given names, the first given name with the remaining given names represented by initials, or initials only, depending on the availability of these name components in the tax records. The full name was always available in the population registers.

Here and in the following sections, we train and evaluate our methods using two ground-truth datasets comprising manually labelled candidate blocks for a sample of tax records from selected municipalities and years (municipality-years). We created two training sets. The first, developed as a pilot study, contained only records from Utrecht in

[1] Originally, 90 municipalities were included in the sample, of which 41 lacked digitised population registers. Initial availability checks had proved misleading because the reported event place referred to the place of birth rather than the place of population registration. Following further investigation, we selected replacement municipalities of similar population size in 1900 and from the same province for 35 of these cases. The problem was particularly pronounced in Limburg, where population registers were rarely digitally indexed or published. Municipalities whose tax data had already been collected were retained in the dataset, although their records could not be linked to the population registers.

1909 (n = 888), while the second contained records from multiple municipality-years (n = 1,400). We manually identified the most likely match within each candidate block, allowing for discrepancies arising from the inherently noisy nature of handwritten historical records. Labelling widows also posed difficulties, as they were frequently recorded under their deceased husbands' names, while maiden names were recorded infrequently.

The quality of our blocking method is high. In Utrecht, where we piloted our methods, the ground-truth match, when present, was ranked first in 96.6% of cases and appeared among the top two candidates in 99.3% of cases. Although Utrecht is a relatively large city, where many people may share the same name, addresses were available, helping to distinguish between otherwise similar records. In the dataset covering multiple municipalities, the scores were lower because, in some municipalities, we had to work with as little as a name. Nevertheless, 92.5% of the best matches appeared among the top three candidates, and this figure rose to 96.2% when the top five candidates were considered.

The second part of the pipeline involves predicting links within these candidate blocks. Each query from the HO tax data is compared with its potential matches in the corresponding candidate block. For this task, we employed a learning-to-rank approach to score candidate pairs, ranking the candidates within each block according to their likelihood of being a match. Specifically, we trained a Gradient Boosted Decision Tree (GBDT) model using the LambdaRank ranking objective (Burges et al. 2006), as implemented in the LightGBM library (Ke et al. 2017). Gradient boosting models can capture complex interactions and non-linear relationships between input features and handle missing values (Rijpma et al. 2020; Fourie et al. 2025). We used the Optuna library to optimise the hyperparameters. For each model, we determined a threshold above which the highest-ranked candidate would be considered a match, balancing precision and recall.

The models are trained on manually labelled data to rank candidates based on a set of features taken directly from, or inferred from, both the tax records and the candidate records, as well as features derived from comparisons between them. These include various similarity metrics for names (e.g. initials, given names, and surnames) and addresses (e.g. street, district, and house number), as well as demographic characteristics such as gender and position within the household.

We trained three distinct models using two datasets. The first model was developed as a pilot study to explore the linkage challenge in depth. We selected Utrecht in 1909 because of the high availability and quality of its data, including full addresses in both the tax registers and the population registers. As Utrecht was the largest city in our sample, achieving accurate linkage there would also be a significant accomplishment in itself. We classified the highest-ranked candidate as a match if its score exceeded a threshold of 0.487. The model performed extremely well, with a test-set precision of 0.988 and recall of 0.992. In evaluating performance, we also used a 'group-correct' statistic, which indicates whether either a correct match was predicted for an individual or the

absence of a true match within the candidate block was correctly identified. The group-correct score for Utrecht in 1909 was 0.989. Given this exceptional performance, we decided to retain this pilot model for Utrecht and for other municipalities in which street addresses were available in both sources.

The second phase of the linkage process built on the lessons learned from the pilot study to develop a procedure and train a model capable of predicting links reliably across the full tax dataset. At this stage, the challenges posed by heterogeneity in both the tax registers and the population registers became apparent. Names and initials were the only features consistently shared between the two data sources. The availability of additional linking information varied considerably between municipalities. We therefore adopted a municipality-by-municipality approach to linkage, adapting the blocking procedure and post-processing to the specific context.

We selected 14 municipality-years with varying levels of data availability, ranging from sparse to rich, and covering different population sizes. Within these municipality-years, we identified common combinations of available linking information, such as initials and surname only, first name and surname, name and neighbourhood, and name and occupation. For this phase, the 1,400 labelled households from the tax registers across these 14 municipality-years were selected using stratified sampling to ensure that challenging cases, such as widows and women, were adequately represented. Note that this oversampling of difficult cases means that the performance metrics reported below probably underestimate the model's true performance on the complete dataset, in which such cases are less frequent.

This model required a substantial number of features to perform well, including semantic distances between strings, conventional string-distance metrics, dummy variables, and various machine-learning inferences, such as predicting gender from names and identifying household positions within the population registers. We also estimated a separate model for widow-headed households, as these cases were exceptionally challenging and hampered learning in the model for non-widow-headed households.

The performance of this model was lower than that of the dedicated Utrecht model. The main reason for this difference was the lack of address information in the population registers of most of the twelve municipalities. Nevertheless, given the inherent sparsity of the original data and the considerable variation in the available features, the model's overall performance remains very good. We select the highest-ranked candidate link whose confidence score exceeds 0.49 for the non-widow model and 0.54 for the widow model. When several plausible alternative links are available, as measured by a standard deviation in predicted probabilities among the top three candidates, we discard the link to maintain a high level of confidence in the remaining linkages. For non-widowed individuals, the model achieves a test-set precision of 0.936 and recall of 0.94329, yielding an F1 score of 0.939 and a group-correct score of 0.914. For widowed individuals, the more challenging group, test-set precision is 0.771 and recall is 0.730, yielding an F1 score of 0.750 and a group-correct score of 0.729.

The resulting predictions are used to create linksets, allowing users to adjust the linkage criteria to suit their research objectives. Assessing the overall quality of the links across the entire dataset is challenging because of the large number of records in both the tax and population register datasets. The current thresholds are optimised for model performance on our ground-truth (GT) datasets. These datasets were designed to reflect the full dataset in terms of the availability of linking features, while placing greater emphasis on challenging records.

Data Description

To describe the resulting dataset, we first compare the final sample with the original sampling design (see Table 1). Our aim was to cover 10% of the Dutch population at ten-year intervals, corresponding to census years, using a stratified sample of municipalities. However, some municipalities in the original sample did not provide sufficient usable records for inclusion. Moreover, the available records in many municipalities did not extend back to 1850.

year	nhh Netherlands	nhh originally sampled	nhh actually sampled	nhh in tax data
1859	687,302	67,500	23,715	12,857
1869	741,303	73,049	37,061	18,518
1879	829,760	79,422	51,965	29,860
1889	932,124	89,975	65,794	36,198
1899	1,053,610	102,918	83,363	47,632
1909	1,208,488	117,691	100,448	68,238
1920	1,415,274	144,846	124,174	108,322

Table 1: Estimated number of households in the Netherlands, and in the municipalities of the original and realised samples, 1859–1920.

As a result, the final municipal sample covers less than the intended 10% of the Dutch population. By the end of the sample period, coverage reaches approximately 9%. However, coverage is considerably lower during the first two decades, as the larger cities enter the dataset only later in the period (Utrecht in 1870, Leiden in 1879, Nijmegen in 1889, and Enschede in 1878). Researchers will need to account for this limitation, for example by using subsampling or extrapolation methods. At the time of writing, we are

unaware of any large Dutch cities with HO tax registers available for earlier years that could improve coverage in this respect.



Figure 5: Coverage of households by the HO tax, 1859–1920.

Beyond the municipalities included in the sample, household coverage is also determined by the coverage of the HO tax itself, which in turn depends on the income threshold. Figure 5 shows how the coverage of the HO tax evolved over time. Around 50–60% of households were covered in 1860, in line with the coverage of other income taxes in the pre-WWI period (Roikonen and Heikkinen 2018). From 1900 onwards, however, coverage improved markedly as incomes rose while the income threshold remained largely unchanged. By 1920, coverage exceeded 90%, which is very high for this period and compares favourably with that of the national income tax introduced in 1915 (Salverda and Atkinson 2007, 460, report 68% coverage in 1919/20).

Two important implications follow. First, analyses should take account of the population below the income threshold and correct for its exclusion where possible. Second, the high coverage of the 1920 data offers a unique opportunity to study the income distribution as a whole at a time when this is often impossible.

Another way to assess the coverage of the data is to examine the types of tax measures reported in the registers (see Table 2). As the HO tax was administered by municipalities, the information recorded varies considerably across registers, even though the tax was intended to target income throughout the country. Tax due was recorded consistently, but in the early decades, particularly in 1859, it was often the only available source of information on income. By 1909 and 1920, gross income, our preferred income measure, had become the most commonly reported measure. As discussed above,

harmonising these different tax measures remains a challenge when working with the dataset.

year	municipalities	households	gross income	taxable income only	tax only	occupation
1859	50	12,857	1,200	0	11,657	6,864
1869	50	18,518	3,365	4,894	10,259	5,643
1879	57	29,860	8,698	8,555	12,607	7,479
1889	57	36,198	22,138	1,164	12,896	10,088
1899	63	47,632	35,786	1,095	10,751	11,036
1909	67	68,238	56,920	6,417	4,901	12,353
1920	66	108,322	100,253	4,635	3,434	63,705

Table 2: Availability of various tax measures in the HIP-NL database.

Occupations provide valuable additional information on the economic structure and income dynamics of this period. They are available for approximately one-third of the households in the sample. However, reporting practices vary across municipalities, so any biases arising from selecting observations based on the availability of occupational information should be carefully examined.

year	individuals	municipalities with any links	linked w. DoB	linked w. DoB & PoB
1859	13,322	22	5,084	3,391
1869	19,013	31	11,342	9,571
1879	30,985	34	14,966	11,939
1889	38,116	30	18,214	15,438
1899	50,293	38	28,935	22,732
1909	71,023	35	41,463	30,847
1920	116,562	31	48,368	30,837

Table 3. Number of individuals by linkage status, showing the availability of date of birth (DoB) and place of birth (PoB) information.

Note: Linkage is performed at the individual level, so the figures reported here are not aggregated to households.

Linking households across tax records to create a panel, and linking individuals to other databases and sources, were key objectives of the HIP-NL project. The first step is to link the tax records to the population registers, which provide crucial information on dates and places of birth. Combined with the names of taxed individuals, these data enable individuals to be uniquely identified in Dutch historical microdata. Although individuals can also be linked within the dataset using less information, these linkage rates provide a useful indication of how successfully individuals can be traced within and across sources.

We link approximately half of the individuals in our dataset to a population register. Most of the attrition is because only about half of the municipalities have population registers digitised to a sufficient extent for linkage. Linkage always provides a date of birth and, for 73% of linked individuals, a place of birth. Linkage rates remain stable over time, although they are somewhat lower in 1859 and 1920. The lower rate in 1920 is largely attributable to the fact that the Utrecht population registers are not available after 1910.

Harmonisation and Imputations

There was considerable variation in the implementation of the HO tax, and the registers differ in the income information they provide. In addition, population coverage and the years and municipalities for which registers survive also vary. In another paper currently in progress, we discuss in detail the steps needed to turn these records into comparable household income distributions. Here, we provide a brief overview of this procedure to highlight potential biases in the raw tax data and the corrections needed to address them.

Although the tax generally applied to households rather than individuals, wives, adult children living at home, and other co-residents with separate incomes could be recorded separately. To harmonise incomes across municipalities, these individual entries should be aggregated at the household level. We suggest using the surname and address, treating multiple individuals with the same surname living at the same address as one household. In 18 registers, this was not possible because addresses were absent and entries were organised alphabetically. Household aggregation affects 5% of the data and has little effect on the resulting inequality measures.

A further issue is that the registers do not report a single, consistent income concept. Gross income is the preferred measure because it is not affected by local deduction rules. However, it is not available for all municipalities, particularly in the earlier years of the tax. Some registers report taxable income instead, which is income after deductions. These deductions could vary between municipalities and could also

depend on household composition. Taxes due are the measure furthest removed from the gross incomes we want to track, as their relationship with gross income depends on local tax rates. Since the HO tax was often progressive rather than a flat tax, all these measures need to be harmonised, even for something as straightforward as calculating inequality measures. Where possible, taxable income or tax due is converted directly into gross income using the local tax rules. Obvious errors in the original records or in the transcription are also corrected where they can be clearly identified.

However, these adjustments do not harmonise incomes across all records. Our preferred solution is to use a model-based procedure to estimate gross income when it is not directly observed. Depending on whether the tax system is expected to be flat or progressive, we use a linear regression model or a gradient boosting model, respectively. These models predict gross income from either tax due or taxable income, using a range of covariates. In our experience, the boosting model is better able to capture the non-linear relationships introduced by deductions, whereas the linear regression model more closely approximates the relatively simple tax systems implied by the sparse registers in which only tax due is reported.

Like most historical income and wealth tax records, the HO registers do not cover the entire population. Most municipalities applied a minimum income threshold, set at subsistence income, below which households were exempt. The observed records therefore cover only the top 50–90% of the local income distribution. Although overall coverage compares favourably with that of other taxes from this period, the lower part of the distribution must be reconstructed to trace income distributions and their evolution over time accurately.

Our approach is to reconstruct the full household income distribution rather than to apply a correction directly to a particular inequality measure (Aitchison and Brown 1957; Alfani and García Montero 2022) or focus exclusively on top income shares (Atkinson et al. 2011). This approach produces a full distribution that can be selected and aggregated in different ways, allowing any required statistic to be calculated. In addition, reweighting this type of data is straightforward.

For each municipality-year, we compare the number of recorded taxpayers with estimates of the total number of households in that municipality-year, derived from the Dutch censuses (Boonstra et al. 2003; Doorn 2019; Meroño-Peñuela et al. 2016). Where census counts of households are unavailable, we estimate the number of households from the total population of the municipality. The number of households to be added is calculated as the difference between the reference population and the number of households recorded in the tax registers.

We fit a censored lognormal distribution to the observed household incomes separately for each municipality-year (Millard 2013). We then draw synthetic households from this estimated distribution, restricting the draws to incomes below the tax threshold

and imposing a lower bound of 40% of the threshold to avoid generating extremely low synthetic incomes.

A final issue is that not every municipality is observed in every benchmark year. For analyses requiring a balanced panel, missing municipality-years need to be estimated. We propose a procedure based on the income distributions of the same municipality in nearby years. Specifically, we interpolate and, where necessary, extrapolate the distribution's percentiles on the log scale.

Overall, of the processing steps described above, reconstructing the lower end of the income distribution has the greatest impact. Although this substantially affects the resulting overall income distributions, it is also an unavoidable step routinely undertaken by economic historians. Similarly, accounting for missing data for large urban centres in 1860–1880 can affect the resulting trends. By contrast, decisions concerning income harmonisation and household aggregation appear to have considerably smaller effects.

Applications of the data

Looking at two applications illustrates the potential of the data assembled by the team, building on its earlier work. The first application involves reconstructing income inequality in the Netherlands between 1860 and 1920, a study that primarily requires the tax data and reference population figures. The second application also draws on the linked data, investigating the income premium or penalty experienced by migrants to the city of Utrecht, whom we can identify using place-of-birth information derived from the population registers.

Inequality

An obvious application of these data is the reconstruction of income inequality in the Netherlands during the crucial period from 1860 to 1920. This period witnessed major economic, political, and demographic changes, yet our current understanding of inequality during this time is based on limited data, primarily the pioneering work of Soltow and Van Zanden (1998) and the broad reconstructions covering this period by Chancel and Piketty (2021: 3037). For the Netherlands, more accurate series based on national income tax data are available only from 1914 onwards (Salverda and Atkinson, 2007; Salverda, 2013).

Here, we briefly present an initial reconstruction of income inequality in the Netherlands using top income shares for the period 1870–1910 and link these estimates to those from the World Inequality Database for the period 1914-1940. This particular application does not require the linked data, but it nevertheless demonstrates HIP-NL's potential for the study of income inequality. The reconstruction presented here builds on the harmonisation procedure outlined in the section above. The resulting inequality series are presented in figure 6 below.

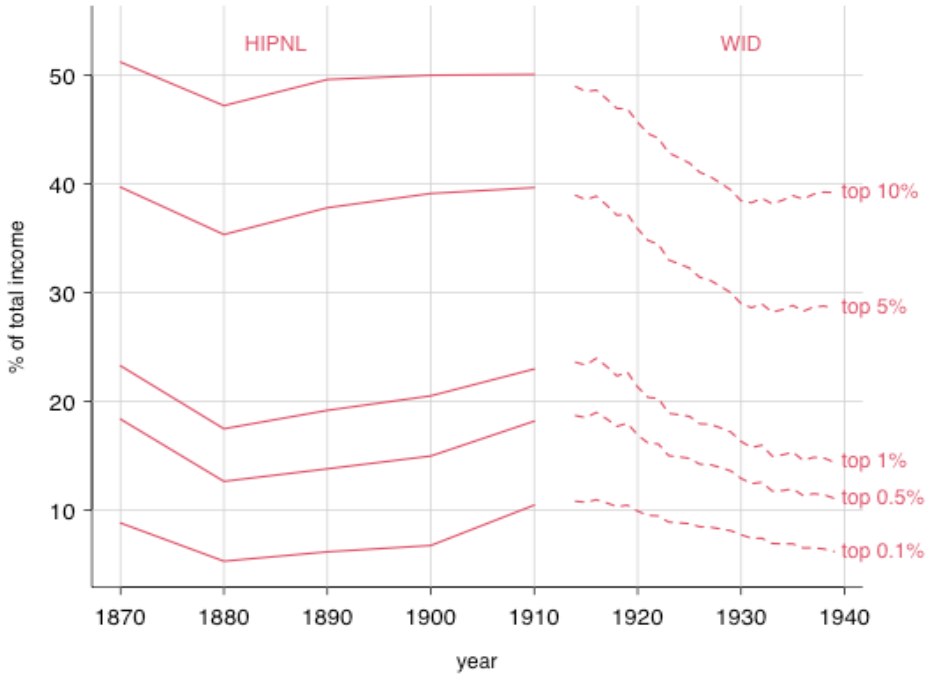


Figure 6. Top income shares in the Netherlands, 1870–1939. Source: 1870–1910, HIP-NL database. 1914–1939: World Inequality Database <https://wid.world>, based on Salverda and Atkinson (2007); Salverda (2013).

The first thing to note is that the income shares at the end of the HIP-NL series line up very well with the later series covered by Salverda and Atkinson based on national income taxes, which gives us confidence in our reconstruction. The period prior to 1914, covered by the HIPNL database, suggests that top income shares in the Netherlands were high and slowly rising from 1880 to 1914. This roughly matches the trends sketched in earlier work based on coarser data (Soltow and Van Zanden; Chancel and Piketty 2021). The new aggregate inequality measures derived from HIP-NL therefore do not upend our understanding of the broad patterns of inequality trends in this period. We see this as encouraging, however, as it means our microdata can largely reproduce these macrotrends and shows strong continuity with later data sources. The next challenge, then, is to test hypotheses about the driving factors of inequality, such as the effects of structural change and migration or democratisation, on the microdata itself.

Migration

The second application focuses on an analysis of the incomes of migrants compared to natives. Research has so far largely relied on occupational titles (e.g., Paping and Pawlowski 2018, Puschmann 2015, Schrover, 2002), an approach that likely obscures considerable income variation within occupational groups. There are few studies

conducted on this topic using income data directly for this period, with much work focusing on later census data in the U.S. (e.g. Boustan 2009). This analysis is relevant to the ongoing debate surrounding the labor market inclusion of migrants, with pessimistic (going back to the Chicago School of Sociology), optimistic (based on the Annales School and the New Economic History), and intermediate perspectives (Lucassen, 2005; Puschmann, 2015) – the latter often differentiating between short-distance and long-distance migrants, as well as stayers versus leavers. The linked HIP-NL data has the potential to bring new precision to this debate.

Here, we present a comparison of migrants and natives in the city of Utrecht for the years 1889 and 1909, based on income data from the taxes, and place of birth data from the linked population registers. Figure 7 illustrates that migrants are over-represented in the middle of the income distribution and under-represented at the bottom and the top.

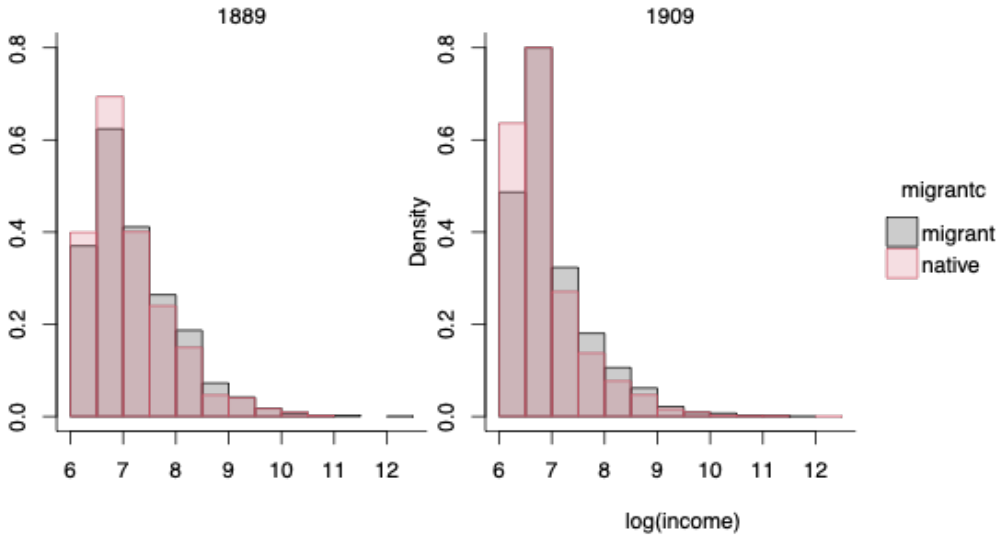


Figure 7: Income distribution of migrants v. natives in Utrecht, 1889 and 1909.

We can also more formally test these observations within a regression framework. The detailed demographic information available in our linked data allows us to adjust these income estimates for age, an important factor if migrants exhibit a different age profile compared to Utrecht natives.

	log(income)	log(income)	log(income)
Migrant	0.102*** (0.008)		
Far migrant		0.136*** (0.009)	
Close migrant		-0.098*** (0.013)	
Urban migrant			0.175*** (0.011)
Rural migrant			0.008 (0.010)
Noble	1.42*** (0.067)	1.43*** (0.071)	1.42*** (0.071)
Year FE	Yes	Yes	Yes
age FE	Yes	Yes	Yes
Observations	26,036	24,113	24,107
R2	0.12121	0.13178	0.13024
Within R2	0.04473	0.05550	0.05387

Table 4: Regression of log(income) on migrant status in Utrecht, 1889 and 1909.

Regression analysis shows that migrants on average have incomes approximately 10% higher than natives (table 4). This income premium is concentrated among urban, long-distance migrants – suggesting selection effects, most likely related to human capital – as opposed to those originating from Utrecht's immediate hinterlands. These findings provide evidence of a positive view of migrants' economic success during this period. We anticipate that the full HIP-NL dataset, with its longitudinal dimension, will further enrich this picture by allowing us to follow migrants over their life courses, and to evaluate whether the successful migrants were immediately successful or needed time to integrate into the labour market.

Conclusions and future research

In this paper, we have outlined the construction of HIP-NL, a database that opens new avenues for research in economic and social history as well as historical demography. By digitizing and linking a nationwide sample of historical municipal tax registers to population registers, HIP-NL makes a wealth of economic, social, and demographic data on inhabitants of the Netherlands between 1850 and 1920 available. Individuals and

households can be linked over time and across generations and, to an extent, across geographical locations. This creates opportunities not only for the study of inequality and living standards, but also income dynamics over the life course and the intergenerational transmission of economic advantage. The data can also be integrated with the broader Dutch microdata infrastructure and potentially linked to income data for later periods.

The two case studies illustrate the analytical potential of HIP-NL. The first confirms high and growing levels of inequality prior to WWI, and is consistent with other estimates in the literature for the Netherlands and with recent estimates for Sweden (Bengtsson & Molinder, 2024). The second case study compares the incomes of migrants and natives in Utrecht. It shows that migrants were not concentrated at the bottom of the income distribution: urban and long distance migrants actually enjoyed an income premium. This finding is consistent with previous studies based on occupational data suggesting that long-distance migrants were positively selected in terms of human capital and benefited from social networks at their destinations (Lucassen, 2004; Puschmann, 2015; Sewell, 1985).

The possibilities for future research extend beyond these initial applications. First, HIP-NL offers opportunities to investigate intragenerational and – once connected to the family relations from the civil registries – intergenerational mobility. Which groups benefited from the massive social, economic and political changes at the end of the nineteenth and start of the twentieth century? To what extent were these developments associated with industrialization, the emergence of the labour movement, or the first steps towards a system of social security?

Second, the geographical information contained in the database makes it possible to investigate the spatial dimensions of income inequality. This includes the breakdown of the income distribution, between the more urbanized and economically developed parts and rural and peripheral regions of the Netherlands? In most municipalities, the HO taxes also provide address-level granularity, allowing us to analyse the degree of concentration of spatial segregation.

Third, the combination of income and demographic information offers new opportunities to investigate gender differences in economic position. Although the available tax records provide limited information on the incomes of married women, the incomes of unmarried and widowed women can be compared both across these groups and with those of men. Which women achieved the highest incomes, where did they live, and what were their family and occupational backgrounds? Which sectors of the economy offered women the best opportunities to earn an independent living, and how did these opportunities change over time? Such analyses would provide new insights into women's economic opportunities in a period when their access to paid employment and independent economic activity was shaped by substantial legal, social, and institutional constraints.

A fourth research avenue concerns partner selection and the relationship between economic position and family formation. Historical research has often emphasized the importance of social homogamy, but the nineteenth and twentieth centuries witnessed changing ideals of marriage and romantic love (Coontz, 2006; Wienholts, et al., 2024). To

what extent did individuals marry into households with higher or lower incomes than those of their own families? Did patterns of economic homogamy vary by occupation, age, and place of residence, and did they change over time? By linking income information to demographic records, HIP-NL could help establish whether economic similarity between partners' families persisted despite broader changes in marriage ideals and social structure.

Fifth, linking income to health and mortality records offers opportunities to investigate the origins of socioeconomic inequalities in health. Some scholars have argued that pronounced health inequalities between rich and poor emerged primarily after the Second World War, when chronic and lifestyle-related diseases became important causes of death (Bengtsson et al., 2020). Some studies, however, have identified socioeconomic gradients in health as early as the nineteenth century, when infectious diseases were major drivers of mortality (e.g. Van Poppel et al., 2009). Because research often relied on occupational status as a proxy for socioeconomic position, HIP-NL offers an opportunity to examine these relationships using more direct measures of income. Further linkage to increasingly available microdata on causes of death, makes it possible to investigate differences in longevity and causes of death by income across the epidemiological transition. Such research would help clarify when socioeconomic inequalities in mortality emerged and how they varied across causes of death. This will be facilitated by the planned extensions of the database with Amsterdam and Maastricht, cities for which historical cause-of-death microdata are available.

Income, however, represents only one dimension of socioeconomic position. Wealth is another important dimension, capturing accumulated assets and economic security rather than current income alone. The Netherlands possesses valuable historical data on property and inheritance, including both detailed inheritance-tax assessments and their summaries for the Netherlands in 1921 (Peeters and De Vicq, 2025; De Vic and Peeters, 2020), which are currently being digitised in a related project. Linking these data to HIP-NL opens up new questions about the relationship between income and accumulated economic resources. To what extent did high incomes translate into substantial wealth? Can we distinguish established self-made from patrimonial wealth (Piketty et al. 2014)? Such analyses would contribute to a more comprehensive understanding of the distribution and persistence of economic power.

The combination of occupational and income data contained in HIP-NL offers a further research avenue in its own right. How did the remuneration of occupations change between 1850 and 1920? Did industrialization and technical change affect the incomes of different occupational groups? What happened to incomes in the agricultural sector during the agricultural crisis and the subsequent restructuring of agricultural production? More broadly, the database provides an opportunity to assess how well-established occupational measures of socioeconomic position, including HISCLASS (Van Leeuwen, & Maas, 2011), HISCAM (Lambert, et al., 2013), and SOCPO (Van de Putte & Miles, 2005), correspond to observed income differences. Comparing their explanatory power and performance across periods and population groups would help establish which measures are most appropriate for different research questions and how much economic heterogeneity remains hidden within occupational categories. We can take this one step

further by developing historical income scores for occupations over this period. In contrast with current measures of occupational status, which capture the social stratification associated with occupations, income scores would provide a measure of occupational economic power based on observed income.

Because HIP-NL provides the full name, address, and date and place of birth for a large share of the recorded household heads, the data to be linked to a range of other historical sources and databases beyond those discussed so far. These include militia registers, which provide information on conscription and physical stature; personnel records from large companies and firms; notarial deeds; and insurance records. Such linkages would enrich the database with additional information on health, working lives, property, and economic security. They would also allow researchers to investigate how income and socioeconomic position related to outcomes and experiences that cannot be reconstructed from tax and demographic records alone.

Taken together, these opportunities show that HIP-NL is more than a resource for reconstructing historical income distributions. By connecting economic information of households and individuals to demographic data and geographical contexts, it provides a basis for investigating how economic resources were distributed, accumulated, and transmitted during a period of profound social and economic transformation. Its integration with existing and future historical data infrastructures will make it possible to examine the relationship between economic inequality and other dimensions of welfare and life chances in greater detail. In doing so, HIP-NL can contribute not only to a more precise measurement of historical economic inequality, but also to a broader understanding of how economic and social inequalities developed and persisted in the Netherlands.

Appendix

Main variables collected for HIP-NL database

Variable	Description
Page number	Register page number.
Place	Municipality place or village name, if the register contains multiple places. (Neighbourhoods/wijken are entered separately).
Reference number	The reference number (*volgnummer*) for each entry.
Neighbourhood	Neighbourhood (*wijk*). Can be entered for the first row only.
House number (wijk)	House number within the neighbourhood, if so organised.
Street	Street name.
House number (street)	House number within the street.
Title	Honorific title (e.g., baron, weduwe, dr, mevr, mej). Uses Dutch titles.
First names	First names, separated by spaces or hyphens if in the original manuscript (e.g., "Johannes" from *Johannes C. van Buren*).
Initials	Initials as stated in the register, separated by spaces or hyphens without periods (e.g., "C" or "J C").
Surname	Surname, including prefixes (e.g., "van Buren").
Suffix	Additional information that does not fit in title or name, such as junior, senior, zoon, etc.
Female	1 if it is very likely a woman, empty otherwise. Assessed based on title, name, or occupational title (widows under their husband's name, e.g., *weduwe gerrit de groot*, are coded 1).
Occupation	Occupation. Some limited standardisation within registers by RAs allowed.
Number of children	Number of children.
Income class	Income class, using the same codes as used in the tax schedule.
Gross Income	Gross income in guilders.
Taxable Income	Taxable income in guilders.
Unspecified Income	Income in guilders when it is unclear from the register whether it concerns gross or taxable income

Variable	Description
Tax	Tax amount in guilders.
Months	Number of months the tax is based on. Entered only if relevant.
Register notes	Any notes found in the register itself.
Entry notes	Notes from the coder

Note on municipal council corrections: In many registers, a corresponding council (*raad*) version of income or tax variables could also exist, representing corrections made by the municipal council to the original entries.

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